

MATH1231 CALCULUS

Chapter 2B

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(2.4) Rational functions and partial fractions

A rational function has the form

$$\frac{p(x)}{q(x)} \quad \text{where } p(x), q(x) \text{ are polynomial functions.}$$

(A polynomial fn of degree n has the form $p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$, for integer $n \geq 0$.)

A rational function is **proper** if the degree of $p(x)$ is less than the degree of $q(x)$

Any rational function can be integrated by **systematic** reduction with **partial fractions**

Quick examples

rational fn? *proper* rational fn?

a.) $\frac{2x+1}{3x^2-4}$

b.) $\frac{\cos x}{x^2+1}$

c.) $\frac{x^2}{x-1}$

d.) $\frac{e^x}{x+1}$

Example

$$I = \int \frac{2x^3 + 7x^2 + 3}{2x + 3} dx$$

Building some intuition

An integral like

$$\int \frac{dx}{x(1-x)}$$

cannot easily be evaluated directly.

However, we could easily do the integrals $\int \frac{1}{x} dx$ and $\int \frac{1}{1-x} dx$.

Can we write $\frac{1}{x(1-x)}$ as a linear combination of $\frac{1}{x}$ and $\frac{1}{1-x}$?

YES!

Partial Fractions

- It can be shown using algebra that every proper rational function $f(x)$ can be written as a unique sum of functions of the form

$$\frac{A}{(x - a)^k} \quad \text{and} \quad \frac{Bx + C}{(x^2 + bx + c)^k},$$

where the quadratic $x^2 + bx + c$ is **irreducible** (i.e., $b^2 - 4c < 0$).

- This sum is called the **partial fractions decomposition** of f .
- Any rational function can be integrated by systematic reduction with partial fractions.

Examples

Some examples of proper rational fractions and their partial-fractions template

$$\frac{1}{(x-2)(x-3)} =$$

$$\frac{3x^2 - 3x + 1}{(x-1)^3} =$$

$$\frac{3x + 2}{(x-1)(x^2 + 2x + 2)} =$$

$$\frac{5x^3 - 3x^2 + 7x - 3}{(x^2 + 1)^2} =$$

Case 1. $q(x) = \alpha(x + a)$ a single linear factor

$$\int \frac{r(x)}{q(x)} dx \quad \deg r(x) < \deg q(x) \Rightarrow r(x) = r = \text{const}$$

$$\begin{aligned} \int \frac{r}{q(x)} dx &= r \int \frac{dx}{\alpha(x - a)} \\ &= \frac{r}{\alpha} \ln |x - a| + C \quad (x \neq a) \end{aligned}$$

Case 2. $q(x) = \alpha(x - a_1)(x - a_2) \dots (x - a_n)$ – distinct linear factors

Write
$$\frac{r(x)}{q(x)} = \frac{A_1}{x - a_1} + \frac{A_2}{x - a_2} + \dots + \frac{A_n}{x - a_n}$$

Then

$$\int \frac{r(x)}{q(x)} dx = \sum_j \int \frac{A_j}{x - a_j} dx$$

How to find A_j ?

Example, to find A_2

multiply both sides of $\frac{r(x)}{q(x)} = \frac{A_1}{x-a_1} + \frac{A_2}{x-a_2} + \dots + \frac{A_n}{x-a_n}$ by $x-a_2$
then

$$\frac{r(x)(x-a_2)}{\alpha(x-a_1)(x-a_2)(x-a_3)\cdots(x-a_n)} = A_1 \frac{x-a_2}{x-a_1} + A_2 \frac{x-a_2}{x-a_2} + \dots + A_n \frac{x-a_2}{x-a_n}$$

Now consider the limit $x \rightarrow a_2$.

$$A_2 = \lim_{x \rightarrow a_2} \left(\frac{r(x) \cancel{(x-a_2)}}{\alpha(x-a_1) \cancel{(x-a_2)} (x-a_3) \cdots (x-a_n)} \right)$$

This technique is called the **Cover-Up Rule**. In effect, we find A_2 by covering up the factor $(x-a_2)$ in the denominator of $\frac{p(x)}{q(x)}$ and substituting $x = a_2$.

Example

$$I = \int \frac{dx}{(x-2)(x-3)}$$

Case 3.

$$q(x) = \alpha(x - a)^k$$

– a power of a linear factor

Write
$$\frac{r(x)}{q(x)} = \frac{A_1}{(x - a)} + \frac{A_2}{(x - a)^2} + \cdots + \frac{A_k}{(x - a)^k}$$

Multiply both sides by $(x - a)^k$ then consider the limit $x = a$ to find A_k

Multiply both sides by $(x - a)^k$ then **differentiate** and consider the limit $x = a$ to find A_{k-1}

Multiply both sides by $(x - a)^k$ then **differentiate twice** and consider the limit $x = a$ to find $A_{k-2} \dots$

Example

$$\int \frac{x^2 - 3}{(x - 1)^3} dx$$

Case 4.

$$q(x) = \alpha(x - a_1)^{k_1}(x - a_2)^{k_2} \dots (x - a_n)^{k_n}$$

– powers of distinct linear factors

Write

$$\frac{r(x)}{q(x)} = \frac{A_{11}}{(x - a_1)} + \frac{A_{12}}{(x - a_1)^2} + \dots + \frac{A_{1k_1}}{(x - a_1)^{k_1}} +$$
$$+ \frac{A_{21}}{(x - a_2)} + \dots + \frac{A_{2k_2}}{(x - a_2)^{k_2}} + \dots$$
$$\dots + \frac{A_{n1}}{(x - a_n)} + \dots + \frac{A_{nk_n}}{(x - a_n)^{k_n}}$$

The constants A_{jk} can be found by multiplying both sides by the denominator and considering special choices for x . This usually involves solving simultaneous equations

Example

$$I = \int \frac{dx}{x(x+1)^3}$$

Sample Class Test Problem

$$I = \int \frac{x+3}{(x+1)(x+2)^2} dx$$

Case 5.

$$q(x) = \alpha(x^2 + bx + c)$$

– irreducible quadratic if $b^2 - 4c < 0$

$$\int \frac{r(x)}{q(x)} dx \quad \text{deg. } r(x) < \text{deg } q(x)$$

w.l.o.g. write

$$r(x) = L(2x + b) + M$$

$$\begin{aligned} \int \frac{r(x)}{q(x)} dx &= \frac{L}{\alpha} \int \frac{2x + b}{x^2 + bx + c} dx + \frac{M}{\alpha} \int \frac{dx}{x^2 + bx + c} \\ &= \frac{L}{\alpha} \ln |x^2 + bx + c| + \frac{M}{\alpha \sqrt{c - \frac{b^2}{4}}} \tan^{-1} \left(\frac{x + \frac{b}{2}}{\sqrt{c - \frac{b^2}{4}}} \right) + C \end{aligned}$$

$$I = \int \frac{dx}{x^2 + bx + c}$$

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + \left(c - \frac{b^2}{4}\right) \quad \text{complete the square}$$

$$= \left(c - \frac{b^2}{4}\right) \left[\left(\frac{\left(x + \frac{b}{2}\right)}{\sqrt{c - \frac{b^2}{4}}} \right)^2 + 1 \right]$$

$$I = \frac{1}{\left(c - \frac{b^2}{4}\right)} \int \frac{dx}{\left[\left(\frac{\left(x + \frac{b}{2}\right)}{\sqrt{c - \frac{b^2}{4}}} \right)^2 + 1 \right]}$$

substitute $\tan \theta = \frac{\left(x + \frac{b}{2}\right)}{\sqrt{c - \frac{b^2}{4}}}$ then $\sec^2 \theta d\theta = \frac{1}{\sqrt{c - \frac{b^2}{4}}} dx$

$$I = \frac{1}{\left(c - \frac{b^2}{4}\right)} \int \frac{\sqrt{c - \frac{b^2}{4}} \sec^2 \theta d\theta}{\tan^2 \theta + 1} = \frac{\theta}{\sqrt{c - \frac{b^2}{4}}} = \frac{1}{\sqrt{c - \frac{b^2}{4}}} \tan^{-1} \frac{\left(x + \frac{b}{2}\right)}{\sqrt{c - \frac{b^2}{4}}}$$

Sample Class Test Problem

Evaluate $I = \int \frac{x}{x^2 - 4x + 13} dx$.

You are given that $\int \frac{du}{u^2 + a^2} = \frac{1}{a} \tan^{-1} \frac{u}{a} + C, \quad a \neq 0$

Example

Evaluate $I = \int \frac{3x^2 - 5x + 3}{(x-1)(x^2 - 2x + 2)} dx$.

You are given that $\int \frac{du}{u^2 + a^2} = \frac{1}{a} \tan^{-1} \frac{u}{a} + C, \quad a \neq 0$

Case 6.

$$q(x) = \alpha(x^2 + bx + c)^k, \quad b^2 - 4c < 0$$

– irreducible quadratic powers

$$\frac{r(x)}{q(x)} = \frac{B_1x + C_1}{x^2 + bx + c} + \frac{B_2x + C_2}{(x^2 + bx + c)^2} + \cdots + \frac{B_kx + C_k}{(x^2 + bx + c)^k}$$

to find constants B_j, C_j multiply by $(x^2 + bx + c)^k$, equate coefs of equal powers of x , solve simultaneous equations.

This yields a sum of integrals $B_jx + C_j = L(2x + b) + M$

$$\int \frac{B_jx + C_j}{(x^2 + bx + c)^j} dx = L \int \frac{2x + b}{(x^2 + bx + c)^j} dx + M \int \frac{dx}{(x^2 + bx + c)^j}$$

the first integral is easy with the substitution $u = x^2 + bx + c$.

$$I = \int \frac{dx}{(x^2 + bx + c)^j} = \frac{1}{(c - \frac{b^2}{4})^j} \int \frac{dx}{\left[\left(\frac{(x + \frac{b}{2})}{\sqrt{c - \frac{b^2}{4}}} \right)^2 + 1 \right]^j}$$

substitute $\tan \theta = \frac{(x + \frac{b}{2})}{\sqrt{c - \frac{b^2}{4}}}$ then $\sec^2 \theta d\theta = \frac{1}{\sqrt{c - \frac{b^2}{4}}} dx$

$$I = \frac{\sqrt{c - \frac{b^2}{4}}}{(c - \frac{b^2}{4})^j} \int \frac{\sec^2 \theta d\theta}{(\tan^2 \theta + 1)^j} = \frac{\sqrt{c - \frac{b^2}{4}}}{(c - \frac{b^2}{4})^j} \int \cos^{2j-2} \theta d\theta$$

General Case.

$$q(x) = \alpha(x + a_1)^{k_1} \cdots (x + a_m)^{k_m} (x^2 + b_1x + c_1)^{\ell_1} \cdots (x^2 + b_nx + c_n)^{\ell_n}$$

$$\begin{aligned} \frac{r(x)}{q(x)} = & \frac{A_{11}}{x + a_1} + \frac{A_{12}}{(x + a_1)^2} + \cdots + \frac{A_{1k_1}}{(x + a_1)^{k_1}} + \cdots \\ & + \frac{A_{m1}}{x + a_m} + \frac{A_{m2}}{(x + a_m)^2} + \cdots + \frac{A_{mk_m}}{(x + a_m)^{k_m}} + \\ & + \frac{B_{11}x + C_{11}}{x^2 + b_1x + c_1} + \frac{B_{12}x + C_{12}}{(x^2 + b_1x + c_1)^2} + \cdots + \frac{B_{1\ell_1}x + C_{1\ell_1}}{(x^2 + b_1x + c_1)^{\ell_1}} + \cdots \\ & + \frac{B_{n1}x + C_{n1}}{x^2 + b_nx + c_n} + \frac{B_{n2}x + C_{n2}}{(x^2 + b_nx + c_n)^2} + \cdots + \frac{B_{n\ell_n}x + C_{n\ell_n}}{(x^2 + b_nx + c_n)^{\ell_n}} \end{aligned}$$

multiply by $(x + a_m)^{k_m}(x^2 + b_n + c_n)^{\ell_n}$, equate coefs of x , solve simultaneous equations

Summary

- ▶ Step 1 – **SIMPLIFY TO MAKE PROPER**
if $\deg p(x) > \deg q(x)$ use long division

$$\frac{p(x)}{q(x)} = s(x) + \frac{r(x)}{q(x)} \quad \deg r(x) < \deg q(x)$$

- ▶ Step 2 – **FACTORIZE THE DENOMINATOR**
factorize $q(x)$ as product of linear factors $(x - a)$
and irreducible quadratic factors $(x^2 + bx + c)$
(irreducible if $b^2 - 4c < 0$).
- ▶ Step 3 – **EXPRESS AS A SUM OF PARTIAL FRACTIONS**
the partial fractions are rational functions of the form $\frac{r(x)}{q(x)}$ whose
denominators contain only linear or quadratic factors of $q(x)$.

(2.5) Further Substitution Techniques

Look for a substitution that will convert the integrand to a rational function and then use the method of partial fractions.

Example: $I = \int \frac{dt}{1 + t^{1/4}}$

Tutorial Example

$$I = \int \frac{x^2 - 1}{(x^2 + 1)\sqrt{1 + x^4}} dx$$

$$u^2 = x^2 + \frac{1}{x^2} \rightarrow I = \frac{1}{\sqrt{2}} \tan^{-1} \frac{\sqrt{x^4 + 1}}{\sqrt{2}x} + C \quad x \neq 0$$

$$y = \frac{x^2 - 1}{x^2 + 1} \rightarrow I = \frac{1}{2\sqrt{2}} \sin^{-1} \left(\frac{x^2 - 1}{x^2 + 1} \right)^2 + D \quad x > 0$$

MAPLE→

$$\frac{\sqrt{1 - ix^2}\sqrt{1 + ix^2} \operatorname{EllipticF}\left(x\left(\frac{1}{2}\sqrt{2} + \frac{1}{2}i\sqrt{2}\right), i\right)}{\left(\frac{1}{2}\sqrt{2} + \frac{1}{2}i\sqrt{2}\right)\sqrt{1 + x^4}} +$$

$$2 \frac{(-1)^{3/4} \sqrt{1 - ix^2}\sqrt{1 + ix^2} \operatorname{EllipticPi}\left(\sqrt[4]{-1}x, i, -\sqrt{-i}(-1)^{3/4}\right)}{\sqrt{1 + x^4}}$$

(2.6) **MAPLE NOTES**

MAPLE excels at integration and partial fractions but it isn't omniscient and it doesn't always simplify well